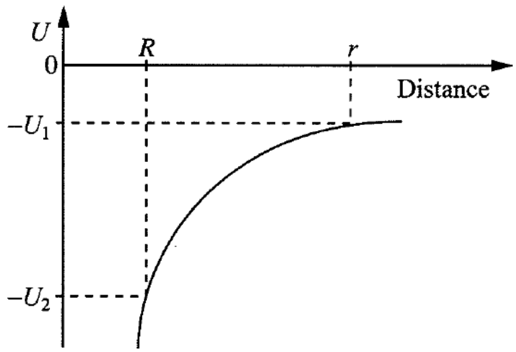
11 The graph below shows the variation of the gravitational potential energy U with the distance from the Earth's centre for a spacecraft. The radius of the earth is R.



If the spacecraft is orbiting the earth in an orbit of radius r, the total energy of the spacecraft is

A $-\frac{1}{2}U_1$ **B** $-\frac{1}{2}U_2$ **C** $-2U_1$ **D** $-2U_2$

Ans: A

Total energy =+U

$$U = -\frac{GMm}{x}$$

$$\frac{GMm}{x^2} = \frac{mv^2}{x}$$

$$K = \frac{1}{2}mv^2 = \frac{1}{2}\frac{GMm}{x}$$

Total energy = U + K $= -\frac{GMm}{x} + \frac{1}{2}\frac{GMm}{x} = -\frac{1}{2}\frac{GMm}{x} = -\frac{1}{2}U$

So at r, total energy = $-\frac{1}{2}U_1$